STA2023 Exam 2 — Ch 4–6 — SIDE 1

■ BLUE = what it means ■ **RED = formula / rule** ■ GREEN = worked question ■ BLACK = steps

SYMBOLS — what they mean

Symbol	Plain English
Σ	"add them all up"
N / n	how many values: N = the whole population, n = a sample
μ ("mu") / x̄ ("x-bar")	the mean (average): μ for a population, x̄ for a sample
σ ("sigma") / s	standard deviation = typical distance of a value from the mean
σ², s²	variance = the SD squared (a middle step)
z	z-score = how many SDs a value sits above (+) or below (−) the mean
P(E)	probability of event E: a number from 0 (impossible) to 1 (certain)
N(E), N(S)	# of outcomes in E; # of outcomes in the whole sample space
E^c	"not E" (the complement)
P(F | E)	probability of F **given** you already know E happened
X / x	X = the random variable (the number an experiment produces); x = one specific value
P(x)	probability that X comes out equal to x
μ_X, σ_X, E(X)	mean, SD of a random variable; E(X) = expected value = same as μ_X
n, p, 1 − p	binomial: # of trials, P(success) on one trial, P(failure)
${}_nC_r$, ${}_nP_r$, n!	ways to choose r of n (order doesn't / does matter); n! = n(n−1)(n−2)···1

Greek letter or capital N → population. Plain letter or small n → sample.
Round probabilities to 4 decimals. Every answer: formula → plug in → arithmetic → boxed answer.

WHICH TOOL? — read the question first

Question says…	Use
table of counts + "or / and / given"	Ch 4 rules on the two-way table
"how many ways"	counting (multiply / ${}_nP_r$ / ${}_nC_r$)
"at least one"	1 − P(none)
table of x and P(x), or counts of outcomes	distribution: valid? μ = Σx·P(x), σ
n trials, probability p, "how many successes"	binomial formula; μ = np, σ = √(np(1−p))
"uniformly distributed between a and b"	rectangle: fraction of the width
"normally distributed, mean μ, SD σ"	z = (x − μ)/σ → table → left / right / between
"what value is the __th percentile / top 10%"	backwards: area → z → x = μ + zσ

EXAM 1 SKILLS THAT COME BACK

- **Mean** = add everything, divide by how many (the balance point). **μ = Σx ÷ N, x̄ = Σx ÷ n**
- **Median** = sort, take the middle (average the two middles if even count); resistant to outliers — the mean is not. **Mode** = most frequent value(s); can be more than one; none if nothing repeats.
- **Standard deviation** = on average, how far values are from the mean. Small SD = values huddle near the mean; big SD = spread out. Why the recipe: the gaps (x − mean) always add to 0, so square them (all positive), average the squares (= variance), then √ to get back to normal units (= SD).

$\sigma^2 = \Sigma(x-\mu)^2 \div N$ (population) · $s^2 = \Sigma(x-\bar{x})^2 \div (n-1)$ (sample) · SD = √variance
"sample" → divide by n − 1 · calculator: σx = population SD, Sx = sample SD

1. Find the mean. 2. Each value − mean. 3. Square each. 4. Add. 5. ÷ N or ÷ (n − 1). 6. √.

Q: "Sample 18, 22, 25, 28, 30: sample variance and SD?" **A:** x̄ = 123/5 = 24.6; gaps −6.6, −2.6, 0.4, 3.4, 5.4; squares 43.56 + 6.76 + 0.16 + 11.56 + 29.16 = 91.20; s^2 = 91.20/4 = **22.8**; s = √22.8 = **4.77**.

- **z-score** = how many SDs away from the mean; + above, − below. Lets you compare different scales: the bigger z did relatively better. **$z = (x-\mu) \div \sigma$** ACT 27 (μ 18, σ 6): z = 1.50 beats SAT 1200 (μ 1000, σ 200): z = 1.00. If lower is better (race times), the smaller z did better. z below −2 or above 2 → unusual.
- **Empirical Rule** (bell-shaped data only): 68% of values within 1 SD of the mean, 95% within 2, 99.7% within 3; the pieces are 34 | 13.5 | 2.35 | 0.15 on each side. μ = 29.8, σ = 6.4: 95% between 29.8 ± 12.8 = 17.0 and 42.6; IQ 85 to 130 = 34 + 34 + 13.5 = 81.5%.
- **Quartiles** cut sorted data into quarters: Q_2 = median, Q_1/Q_3 = medians of the lower/upper halves (odd n → leave the median out). **IQR = $Q_3 - Q_1$; lower fence = Q_1 − 1.5·IQR, upper fence = Q_3 + 1.5·IQR; outlier = below the lower fence or above the upper fence**

CH 4 — PROBABILITY

What probability is

- **Probability** = the long-run fraction of times something happens (flip a coin a million times → about half heads → 0.5). **0 ≤ P(E) ≤ 1; all outcomes together add to 1.** **Unusual event** = probability less than 0.05.
- **Sample space S** = the list of everything that could happen; an **event** = any part of that list you care about. Two coin flips: S = {HH, HT, TH, TT}.
- **Classical** (all outcomes equally likely — dice, cards, coins): just count. **P(E) = N(E) ÷ N(S)** P(two heads) = 1/4 = 0.25; P(at least one tail) = 3/4.
- **Empirical** (from data you collected): how often it happened. **P(E) ≈ frequency ÷ total trials** 22 of 200 carpool → 0.11; 5 walk → 0.025 < 0.05 → unusual.
- **Subjective** = an educated guess when there's no data (chance of a recession).
- **Probability model** = list of outcomes with probabilities; valid only if each is 0–1 and they add to 1. {A 0.40, B 0.20} → not valid (sum 0.60); a −0.10 → not valid.
- **Disjoint** (= mutually exclusive) = no outcomes in common, can't both happen (a card can't be a king AND a queen), so P(E and F) = 0. **Independent** = E happening does not change the probability of F (two coin flips; two different drivers). **Disjoint is NOT independent:** if E and F are disjoint and both possible, knowing E happened means F didn't → dependent.

The rules — spot the keyword

NOT: $P(E^c) = 1 - P(E)$
OR: P(E or F) = P(E) + P(F) − P(E and F) (disjoint → the overlap is 0, just add)
AND, independent: P(E and F) = P(E) · P(F)
AND, general: P(E and F) = P(E) · P(F | E)
GIVEN: $P(F \mid E) = \frac{P(E \text{ and } F)}{P(E)} = \frac{N(E \text{ and } F)}{N(E)}$
AT LEAST ONE: P(at least 1) = 1 − P(none)
Independent? check P(F | E) = P(F). Equal → independent; different → dependent

- **Why subtract in OR:** adding P(E) + P(F) counts the overlap twice, so take it out once.
- **Why "given" changes things:** knowing E shrinks the world to only the cases where E happened, so you divide by E's total, not the grand total. P(6) = 1/6, but P(6 | even) = 1/3 because only {2, 4, 6} remain.
- **"At least one"** has many cases, so compute the one case you don't want (none) and subtract from 1.
- **Independence rule of thumb:** sampling without replacement from a big population — if the sample is under 5% of the population, treat the picks as independent.

Q: "P(speeder pulled over) = 0.8, P(ticket | pulled over) = 0.9. P(pulled over AND ticket)?" **A:** 0.8 × 0.9 = **0.72**
Q: "P(24-yr-old male survives the year) = 0.9986. P(two both survive)? P(at least one of 1000 dies)?" **A:** 0.9986^2 = **0.9972**; $1 - 0.9986^{1000} = 1 - 0.2464$ = **0.7536**
Q: "One card: P(king or diamond)?" **A:** 4/52 + 13/52 − 1/52 = 16/52 = **0.3077** (the king of diamonds was counted twice). "P(king or queen)?" disjoint → 4/52 + 4/52 = 0.1538
Q: "52% played the lottery; P(has not)?" **A:** 1 − 0.52 = **0.48**
Q: "Pick 2 of 4 friends (A, B, S, D): P(A and B)? P(B goes)?" **A:** S = {AB, AS, AD, BS, BD, SD} → 1/6; 3/6 = 0.5
Q: "Are 'widowed' and 'female' independent? P(widowed) = 0.0552, P(widowed | female) = 11.5/138.1 = 0.0833." **A:** 0.0833 ≠ 0.0552 → **dependent**

Two-way (contingency) table ★Q1

A table of counts split two ways (status × gender). Probabilities are just cells or totals divided by the right denominator.

1. Add row totals and column totals; grand total = N.
2. "is A" → A's total ÷ N.
3. "A **or** B" → P(A) + P(B) − P(both) (the overlap cell).
4. "A **and** B" → the one cell ÷ N.
5. "A **given** B" → cell ÷ B's total (divide by the GIVEN group only).

Q: "Employment status of 78 residents: Full-time M 22 / F 18; Part-time M 6 / F 10; Unemployed M 3 / F 2; Not in labor force M 5 / F 12. Males = 36, Females = 42." Row totals: full-time 40, part-time 16, unemployed 5, not in labor force 17.
(a) P(male) = 36/78 = **0.4615**
(b) P(part-time) = 16/78 = **0.2051**
(c) P(male OR unemployed) = 36/78 + 5/78 − 3/78 = 38/78 = **0.4872**
(d) P(female AND not in labor force) = 12/78 = **0.1538**
(e) P(part-time | female) = 10/42 = **0.2381**

Counting — "how many ways"

- A **sequence of choices** (first this, then that): multiply the number of options at each step. 3-letter airport codes: 26·26·26 = 17,576; chair, vice-chair, secretary from 14 people: 14·13·12 = 2,184.
- **Permutation** = picking r of n where ORDER MATTERS (1st/2nd/3rd place, arranging in a line). **${}_nP_r = \frac{n!}{(n-r)!}$** ${}_7P_5$ = 7!/2! = 2,520; 10 runners, top 3: ${}_{10}P_3 = 10 \cdot 9 \cdot 8 = 720$; 7 schools in order: 7! = 5,040.
- **Combination** = picking r of n where order DOESN'T matter (a committee, a hand of cards, a sample). **${}_nC_r = \frac{n!}{r!(n-r)!}$** ${}_4C_2 = 24/(2 \cdot 2) = 6$; samples of 4 from 20: ${}_{20}C_4$ = (20·19·18·17)/24 = 4,845; P(4 cards all same rank) = $13/{}_{52}C_4$ = 13/270,725 ≈ 0.00005.

CH 5 — DISCRETE RANDOM VARIABLES

- **Random variable X** = the number that comes out of a chance experiment (games in a series, profit from a bet, # who say yes). **Discrete** = you count it (0, 1, 2, …: number of pieces, people, heads); **continuous** = you measure it (time, mass, temperature — any decimal).
- **Probability distribution** = a table of every value X can take with its probability (or a graph / formula).

VALID only if (1) every P(x) is between 0 and 1 AND (2) Σ P(x) = 1
Mean (expected value): $\mu_X = E(X) = \Sigma [\, x \cdot P(x) \,]$
SD: $\sigma_X = \sqrt{\Sigma [\, (x-\mu)^2 \cdot P(x) \,]}$ shortcut $\sqrt{\Sigma [\, x^2 \cdot P(x) \,] - \mu^2}$

- **Mean of a distribution** = a weighted average: each value counts as much as its probability. It's the long-run average per trial ("over many games you lose 8¢ a game on average").
- **SD of a distribution** = same idea as before (typical distance from the mean) but each squared gap is weighted by its probability instead of averaged.

1. From counts: P(x) = count ÷ total. Check the P(x) add to 1.
2. Mean: multiply each x by its P(x); add.
3. SD: (x − μ)² for each x; times P(x); add; √.
4. Graph: a vertical line (or bar) above each x, height = P(x); x across, P(x) up.

Q: "Is this a valid discrete probability distribution? x: 0, 1, 2, 3, 4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01." **A:** each P(x) is 0–1 ✓, but 0.34 + 0.21 + 0.13 + 0.04 + 0.01 = 0.73 ≠ 1 → **NOT valid**.

Q: "Games played in the Stanley Cup: x = 4 (20 times), 5 (20), 6 (26), 7 (17). (a) Build the distribution. (b) Graph it. (c) Mean and interpret. (d) SD."
(a) total 83 → P(4) = 20/83 = 0.2410, P(5) = 0.2410, P(6) = 26/83 = 0.3133, P(7) = 17/83 = 0.2048 (sum ≈ 1 ✓)
(b) bars at 4, 5, 6, 7 with heights 0.2410, 0.2410, 0.3133, 0.2048
(c) μ = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048) = 0.9640 + 1.2050 + 1.8798 + 1.4336 = **5.48 games** → over many series the Cup lasts about 5.48 games on average
(d) $(4-5.48)^2(0.2410) = 2.1904(0.2410) = 0.5279$ · $(5-5.48)^2(0.2410) = 0.2304(0.2410) = 0.0555$ · $(6-5.48)^2(0.3133) = 0.2704(0.3133) = 0.0847$ · $(7-5.48)^2(0.2048) = 2.3104(0.2048) = 0.4732$ → $\sigma^2 = 1.1413$ → σ = √1.1413 = **1.07 games**

Q: "Chuck-a-Luck profit: −$1 (0.5787), $1 (0.3472), $2 (0.0695), $3 (0.0046). Mean? SD?" **A:** μ = −0.5787 + 0.3472 + 0.1390 + 0.0138 = **−$0.08** (lose 8¢/game); σ^2 = 0.4912 + 0.4040 + 0.3003 + 0.0436 = 1.2391 → σ = **$1.11**.
Q: "Insurance: +$100 with 0.9986, −$9,900 with 0.0014. Expected profit?" **A:** 99.86 − 13.86 = **$86** per policy → keep selling (positive).

STA2023 Exam 2 — Ch 4–6 — SIDE 2

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CH 5 (cont.) — BINOMIAL

Binomial experiment = the same yes/no trial repeated: (1) a FIXED number of trials n, (2) trials independent, (3) only two outcomes each time (success / failure), (4) the SAME p every trial. X = how many successes. Not binomial if n isn't fixed ("keep going until…") or picks are without replacement from a small group (p changes).

$P(x) = {}_nC_x \cdot p^x \cdot (1-p)^{n-x}$ $\quad {}_nC_x = \frac{n!}{x!(n-x)!}$
$\mu = n \cdot p$ $\quad \sigma = \sqrt{n \cdot p \cdot (1-p)}$ $\quad$ Excel =BINOM.DIST(x, n, p, FALSE) exactly; TRUE = P(X ≤ x)
Unusual count (when np(1 − p) ≥ 10): x below μ − 2σ or above μ + 2σ

Read the formula as: (number of ways to place the x successes among the n trials) × (probability of the x successes) × (probability of the n − x failures).

Words	Compute
exactly x	P(x)
fewer than k / less than k	P(0) + … + P(k − 1)
at most k / no more than k / ≤ k	P(0) + … + P(k)
more than k / > k	1 − [P(0) + … + P(k)]
at least k / ≥ k	1 − [P(0) + … + P(k − 1)]
between a and b (inclusive)	P(a) + … + P(b)

Q: "60% of U.S. women say the minimum driving age should be 18. Random sample of 10. (a) Exactly 3? (b) Fewer than 3? (c) Between 1 and 3 inclusive? (d) Sample of 200: expected number and SD?"
n = 10, p = 0.60, 1 − p = 0.40
(a) $P(3) = {}_{10}C_3(0.6)^3(0.4)^7 = 120 \cdot 0.216 \cdot 0.0016384 =$ **0.0425**
(b) $P(0) = (0.4)^{10} = 0.0001$; $P(1) = 10(0.6)(0.4)^9 = 0.0016$; $P(2) = 45(0.36)(0.4)^8 = 0.0106$ → P(X < 3) = **0.0123**
(c) P(1) + P(2) + P(3) = 0.0016 + 0.0106 + 0.0425 = **0.0547**
(d) μ = 200(0.60) = **120**; σ = √(200 · 0.60 · 0.40) = √48 = **6.93**. "Would 130 be unusual?" 120 ± 2(6.93) = 106.1 to 133.9 → 130 is inside → **not unusual**.

Q: "Binomial or not? (1) Roll a pair of dice 10 times, count 7s. (2) Select flights until 10 late ones are found. (3) Pick 4 of 30 students, count females." **A:** (1) YES: n = 10, p = 6/36 = 1/6. (2) NO: n not fixed. (3) NO: without replacement from 30 (4/30 = 13% > 5%) → not independent.
Q: "On-time rate 0.70, 4 flights: distribution?" **A:** $P(0) = (0.3)^4 = 0.0081$ · $P(1) = 4(0.7)(0.3)^3 = 0.0756$ · $P(2) = 6(0.49)(0.09) = 0.2646$ · $P(3) = 4(0.343)(0.3) = 0.4116$ · $P(4) = (0.7)^4 = 0.2401$
Q: "Free throws 95%, 3 shots: P(all 3)? P(less than 3)? mean & SD?" **A:** $0.95^3 =$ **0.8574**; 1 − 0.8574 = **0.1426**; μ = 3(0.95) = **2.85**; σ = √(3·0.95·0.05) = **0.38**
Q: "72% prefer chocolate, n = 10: exactly 8? fewer than 3? at least 3?" **A:** $45(0.72)^8(0.28)^2 =$ **0.2548**; P(0)+P(1)+P(2) = **0.0010**; 1 − 0.0010 = **0.9990**

CH 6 — CONTINUOUS & NORMAL

Area = probability

- A continuous variable has infinitely many values, so you can't list them. Instead, probability = **AREA under a curve** over an interval. The curve is the **probability density function (pdf)**: never below 0, total area = 1.
- **P(X = one exact value) = 0** (a single point has no width), so P(a < X < b) = P(a ≤ X ≤ b).

Uniform distribution ★Q5

Uniform on [a, b] = a flat rectangle: every stretch of the same length is equally likely (friend is as likely to be 3–4 min late as 25–26 min late). Height = 1/(b − a) so the rectangle's area is 1. Probability of an interval = its share of the total width.

height = $\frac{1}{b-a}$ $\quad P(c \le X \le d) = \frac{d-c}{b-a}$ $\quad$ value with area P to its LEFT: x = a + P·(b − a)

Q: "Daytime temperature X is uniformly distributed between 70°F and 100°F. (a) Sketch/describe the pdf. (b) P(75 ≤ X ≤ 85)? (c) P(X < 80)?"
(a) rectangle from 70 to 100, height 1/(100 − 70) = 1/30 (area 30 × 1/30 = 1)
(b) (85 − 75)/30 = 10/30 = **0.3333** **(c)** (80 − 70)/30 = 10/30 = **0.3333**

Q: "Friend is 0–30 min late, uniform. P(10 to 20 min late)? 20% chance they arrive within the next ___ min?" **A:** 10/30 = 0.3333; 0 + 0.20(30) = **6 min** (70–100°F: coolest 20% → 70 + 0.20(30) = 76°F). "Random number 0–10: P(between 2 and 5)? P(exactly 4.35)?" **A:** 3/10 = 0.3; **0**.

Normal distribution

Normal = the bell curve: symmetric around μ, most values near the middle, fewer far out. Every normal curve has the same shape — μ slides it left/right, σ makes it wider (big σ) or narrower (small σ).

Properties: (1) symmetric about μ; (2) mean = median = mode = μ, the highest point; (3) inflection points at μ ± σ; (4) total area = 1; (5) tails get closer to the axis but never touch; (6) Empirical Rule: 68% within 1σ, 95% within 2σ, 99.7% within 3σ; (7) half the area (0.5) lies on each side of μ.

Standard normal Z = the normal curve with μ = 0, σ = 1. **The trick of the chapter:** you can't have a table for every μ and σ, so convert your value to a z-score and use ONE table (Table V), which gives the area to the LEFT of z.

$z = \frac{x-\mu}{\sigma}$ (round z to 2 decimals) $\quad x = \mu + z \cdot \sigma$
LEFT of x ("less than / below / shorter"): table value
RIGHT of x ("more than / above / at least"): 1 − table value
BETWEEN x_1 and x_2: (left of z_2) − (left of z_1) (bigger area − smaller area)
Negative z: left of −z = 1 − (left of +z)

1. Draw a bell, write μ under the peak, mark the x value(s), shade the region asked.
2. z = (x − μ)/σ for each x, 2 decimals.
3. Table: row = first two digits (1.6), column = last digit (.07) → area to the LEFT.
4. Left → done. Right → 1 − area. Between → subtract the two lefts.

Q: "(a) Area to the left of z = −1.10. (b) Area between z = −0.60 and z = 1.45. (c) X normal, μ = 80, σ = 10; P(70 < X < 95)?"
(a) table → **0.1357**
(b) left of 1.45 = 0.9265; left of −0.60 = 0.2743 → 0.9265 − 0.2743 = **0.6522**
(c) z = (70 − 80)/10 = −1.00 → 0.1587; z = (95 − 80)/10 = 1.50 → 0.9332 → 0.9332 − 0.1587 = **0.7745**

Q: "Wechsler IQ: μ = 100, σ = 15. (a) P(above 125)? (b) P(below 90)? (c) Proportion between 110 and 140? (d) Score at the 98th percentile?"
(a) z = (125 − 100)/15 = 1.67 → left 0.9525 → right = 1 − 0.9525 = **0.0475**
(b) z = (90 − 100)/15 = −0.67 → **0.2514**
(c) z = 0.67 → 0.7486; z = (140 − 100)/15 = 2.67 → 0.9962 → 0.9962 − 0.7486 = **0.2476**
(d) area to the left = 0.98 → closest in table 0.9798 → z = 2.05 → x = 100 + 2.05(15) = **130.75 ≈ 131**

Q: "SAT math μ = 516, σ = 116: P(less than 500)? P(more than 600)? P(between 525 and 550)?" **A:** z = −0.14 → **0.4443**; z = 0.72 → 1 − 0.7642 = **0.2358**; z = 0.08 and 0.29 → 0.6141 − 0.5319 = **0.0822**
Q: "Cholesterol μ = 180, σ = 36.2: P(above 200)?" **A:** z = 0.55 → 1 − 0.7088 = **0.2912**
Q: "Pregnancy μ = 266, σ = 16: P(259 days or less)?" **A:** z = −0.44 → **0.3300**
Q: "Heights μ = 38.72, σ = 3.17: P(shorter than 35)?" **A:** z = −1.17 → **0.1210**
Q: "IQ: P(outside 70 to 130)?" **A:** z = ±2.00 → 0.0228 + 0.0228 = **0.0456**

Backwards — find x from a percent / percentile

Now the area is GIVEN and you want the value. Reverse the steps: area → z (from inside the table) → x = μ + zσ. "k-th percentile" = the value with k% of the data below it, so area to the left = k/100 (5th → 0.05, 20th → 0.20, 98th → 0.98).

1. Draw & shade; turn the words into an area to the LEFT ("top 10%" → left area 0.90; "bottom 5%" → 0.05; "middle 80%" → 0.10 and 0.90).
2. Find that area INSIDE the table (closest value); read z from the row and column. Exactly halfway between two entries → average the two z's (0.95 → 1.645, 0.995 → 2.575).
3. x = μ + z·σ.

Percentile	z	Percentile	z
1st	−2.33	99th	2.33
2nd	−2.05	98th	2.05
2.5th	−1.96	97.5th	1.96
5th	−1.645	95th	1.645
10th	−1.28	90th	1.28
20th	−0.84	80th	0.84
25th	−0.67	75th	0.67

Middle 80% ↔ z = ±1.28 · middle 90% ↔ ±1.645 · middle 95% ↔ ±1.96 · middle 99% ↔ ±2.575

Q: "Heights of 3-yr-old girls μ = 38.72, σ = 3.17. (a) Height at the 20th percentile? (b) Heights that separate the middle 80%?"
(a) area 0.20 → z = −0.84 → x = 38.72 + (−0.84)(3.17) = **36.06 in**
(b) z = ±1.28 → 38.72 − 1.28(3.17) = **34.66** and 38.72 + 1.28(3.17) = **42.78 in**
Q: "IQ μ = 100, σ = 15: top 10%? bottom 5%? middle 90%?" **A:** left area 0.90 → z = 1.28 → **119.2**; z = −1.645 → **75.3**; z = ±1.645 → **75.3 to 124.7**

Interpreting an area · Is it normal?

- "Area to the right of 200 is 0.2903" means BOTH: the **proportion** of the population above 200 is 0.2903, AND the **probability** that one randomly chosen person is above 200 is 0.2903.
- **Sketch:** bell, μ at the peak, ticks at μ ± σ, μ ± 2σ, μ ± 3σ, mark x, shade, write the area in the shaded part.
- **Assessing normality:** histogram should be bell-shaped, one peak, roughly symmetric, few outliers; a **normal probability plot** should be close to a straight line. Restaurant A (tall on the left, long right tail) → skewed, not normal; Restaurant B (symmetric hump) → normal. Greyhound times near a line → approximately normal.

Standard normal table — area to the LEFT of z (entries 0.xxxx; negative z → 1 − value)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	5000	5040	5080	5120	5160	5199	5239	5279	5319	5359
0.1	5398	5438	5478	5517	5557	5596	5636	5675	5714	5753
0.2	5793	5832	5871	5910	5948	5987	6026	6064	6103	6141
0.3	6179	6217	6255	6293	6331	6368	6406	6443	6480	6517
0.4	6554	6591	6628	6664	6700	6736	6772	6808	6844	6879
0.5	6915	6950	6985	7019	7054	7088	7123	7157	7190	7224
0.6	7257	7291	7324	7357	7389	7422	7454	7486	7517	7549
0.7	7580	7611	7642	7673	7704	7734	7764	7794	7823	7852
0.8	7881	7910	7939	7967	7995	8023	8051	8078	8106	8133
0.9	8159	8186	8212	8238	8264	8289	8315	8340	8365	8389
1.0	8413	8438	8461	8485	8508	8531	8554	8577	8599	8621
1.1	8643	8665	8686	8708	8729	8749	8770	8790	8810	8830
1.2	8849	8869	8888	8907	8925	8944	8962	8980	8997	9015
1.3	9032	9049	9066	9082	9099	9115	9131	9147	9162	9177
1.4	9192	9207	9222	9236	9251	9265	9279	9292	9306	9319
1.5	9332	9345	9357	9370	9382	9394	9406	9418	9429	9441
1.6	9452	9463	9474	9484	9495	9505	9515	9525	9535	9545
1.7	9554	9564	9573	9582	9591	9599	9608	9616	9625	9633
1.8	9641	9649	9656	9664	9671	9678	9686	9693	9699	9706
1.9	9713	9719	9726	9732	9738	9744	9750	9756	9761	9767
2.0	9772	9778	9783	9788	9793	9798	9803	9808	9812	9817
2.1	9821	9826	9830	9834	9838	9842	9846	9850	9854	9857
2.2	9861	9864	9868	9871	9875	9878	9881	9884	9887	9890
2.3	9893	9896	9898	9901	9904	9906	9909	9911	9913	9916
2.4	9918	9920	9922	9925	9927	9929	9931	9932	9934	9936
2.5	9938	9940	9941	9943	9945	9946	9948	9949	9951	9952
2.6	9953	9955	9956	9957	9959	9960	9961	9962	9963	9964
2.7	9965	9966	9967	9968	9969	9970	9971	9972	9973	9974
2.8	9974	9975	9976	9977	9977	9978	9979	9979	9980	9981
2.9	9981	9982	9982	9983	9984	9984	9985	9985	9986	9986
3.0	9987	9987	9987	9988	9988	9989	9989	9989	9990	9990
3.1	9990	9991	9991	9991	9992	9992	9992	9992	9993	9993
3.2	9993	9993	9994	9994	9994	9994	9994	9995	9995	9995
3.3	9995	9995	9995	9996	9996	9996	9996	9996	9996	9997
3.4	9997	9997	9997	9997	9997	9997	9997	9997	9997	9998

Reading it: z = 1.67 → row **1.6**, column **.07** → 0.9525. z = −1.67 → 1 − 0.9525 = 0.0475. **Backwards:** area 0.98 → closest entry 0.9798 → row 2.0, column .05 → z = 2.05.

DON'T LOSE POINTS

- "Or" → subtract the overlap once. "And" → the one cell. "Given" → divide by the given group, not the grand total.
- Check Σ P(x) = 1 before calling a distribution valid; probabilities must be 0–1.
- Binomial μ = np, σ = √(np(1−p)) — don't forget the √. Mean of a distribution is Σ x·P(x), not the average of the x's.
- "Fewer than 3" stops at 2; "at least 3" = 1 − P(≤ 2); "at most" includes the number.
- Uniform: probability = length ÷ total width; never use z. Normal: ALWAYS convert to z (2 decimals); table = area to the LEFT.
- "Above / more than" → 1 − left. "Between" → big area minus small area. Negative z → 1 − (positive entry).
- Percentile → area → z (from inside the table) → x = μ + zσ. "Top 10%" means left area 0.90.
- Round probabilities to 4 decimals; write the formula every time.